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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

COMMUNICATION PROTOCOL

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Write the difference between static and dynamic IP addresses?	L2	CO1	**02**
	b.	Define the role of the Application Layer in the TCP/IP protocol.	L2	CO2	**02**
	c.	Describe the features of Bluetooth Low Energy (BLE).	L2	CO3	**02**
	d.	Describe the concept of 6LoWPAN in IoT.	L2	CO4	**02**
	e.	How MQTT is it applied in IoT communication?	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Examine the provider, changes, security, device tracking, stability, and cost for Static IP address vs Dynamic IP address.	L4	CO1	**05**
		OR			
	b.	Define DNS. How does it help in the functioning of the Internet?	L2	CO1	**05**
3.	**a.**	Explain how TCP and UDP ports work in the context of the Application Layer protocols.	L2	CO1	**05**
		OR			
	b.	Illustrate the concept of Network Layer and their functions.	L2	CO2	**05**
4.	**a.**	Examine the concept of Web socket and when it can be used.	L2	CO2	**05**
		OR			
	b.	Differentiate the TCP/IP stack and the OSI reference model.	L3	CO2	**05**
5.	**a.**	Explain how WirelessHART works.	L2	CO3	**05**
		OR			
	b.	Discuss the Zigbee architecture and its layers	L2	CO3	**05**
6.	**a.**	Illustrate the concept of IPv6 over Networks of Resource-constrained Nodes 6lo.	L2	CO4	**05**
		OR			
	b.	Describe the concept of IPv6 over G.9959.	L2	CO4	**05**

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7.	**a.**	With neat sketch of CoAP architecture, explain in detail.	L2	CO5	**05**
		OR			
	b.	Discuss the concept of Extensible Messaging and Presence Protocol architecture.	L2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss how DNS works? Examine different sections of DNS	L2	CO1	**10**
		OR			
	b.	With neat sketch of MAC address format explain why the MAC address should be unique in the LAN network.	L3	CO1	**10**
9.	**a.**	Outline concept of Message Queue Telemetry Transport.	L2	CO2	**10**
		OR			
	b.	Illustrate the concept of Physical Layer and their functions	L2	CO2	**10**
10.	**a.**	With neat sketch of Z-Wave System Architecture, explain two broad categories of Z-wave network	L2	CO3	**10**
		OR			
	b.	Summarize the concept, the Long Term Evolution Advanced (LTE-A)	L2	CO3	**10**
11.	**a.**	Explain the Z-Wave in IoT Data Link Protocol. How does Z-Wave Protocol Work?	L2	CO3	**10**
		OR			
	b.	Illustrate the concept, Network layer Encapsulation Protocols of 6LoWPAN.	L2	CO4	**10**
12.	**a.**	Describe the concept of IPv6 over Bluetooth Low Energy.	L2	CO4	**10**
		OR			
	b.	Discuss the Routing Protocol for Low-Power and Lossy Networks routing metrics.	L2	CO4	**10**
13.	**a.**	Examine how data will transfer from client to server in HTTP message transactions.	L4	CO5	**10**
		OR			
	b.	Describe the concept of Interconnection of Heterogeneous Data links in IoT management protocol.	L4	CO5	**10**